

6. Clustering with Python

1. What?

During this unit, I learned how to perform **K-means clustering using Scikit-learn in Python**. I explored how clustering can group similar data points without predefined labels, making it a powerful unsupervised learning technique.

I learned several important concepts:

- **K-means clustering** and how it partitions data into k clusters based on similarity.
- The use of the **Silhouette Score** as a metric for evaluating clustering quality in unsupervised learning by measuring how well samples fit within their assigned clusters.
- The calculation of **Sum of Squared Errors (SSE)** during K-means training and its use in the **Elbow Method** to determine an appropriate number of clusters.
- The use of a **Confusion Matrix (CM)** to compare cluster assignments with known class labels when evaluating clustering results.
- The application of **Principal Component Analysis (PCA)** to reduce high-dimensional data and visualize cluster results in two or three dimensions.
- A practical tutorial involving **K-means clustering on the Iris dataset**, whose solution was included in Unit 6 of the e-portfolio.

2. So What?

This learning experience helped me understand the complete workflow of an unsupervised machine learning project, from data preprocessing and model training to evaluation and visualization. Unlike supervised learning, clustering does not rely on target labels, which makes assessing model quality more challenging.

The tutorial demonstrated that selecting the correct number of clusters is critical for obtaining meaningful results. Using SSE and the Elbow Method provided a systematic way to estimate the optimal value of k , while the Silhouette Score offered additional evidence of cluster separation and cohesion.

The PCA visualization was particularly valuable because it transformed abstract numerical results into an interpretable graphical representation. Observing the Iris dataset clusters in a reduced feature space improved my understanding of how different flower species relate to each other and how well K-means can separate them.

I also discovered limitations in the use of confusion matrices for clustering evaluation. Unlike supervised classification, cluster labels generated by K-means are assigned arbitrarily. Consequently,

a confusion matrix cannot be interpreted directly unless the cluster labels are first matched to the corresponding real classes. This highlighted the importance of understanding the underlying methodology rather than relying solely on evaluation outputs.

Overall, the exercise reinforced my understanding of unsupervised learning, model evaluation techniques, and the importance of visualization when interpreting complex datasets.

3. Now What?

Going forward, I plan to deepen my knowledge of clustering techniques and unsupervised learning. I would like to:

- Apply clustering to larger and more complex real-world datasets to gain practical experience.
- Investigate additional clustering evaluation metrics, including the **Calinski-Harabasz Index** and **Davies-Bouldin Score**, introduced in Unit 6 Seminar.
- Improve my ability to communicate clustering results through visualizations and data storytelling techniques.